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Abstract

Social networks are the most used method of sharing information and interacting with other people. Major websites like StackOverflow, Reddit, and GitHub are major hubs for information sharing with each designed to incentivize, track, and measure sharing to increase the overall contributions of everyone who visits their sites. Attackers are always developing new tactics to find their way into secure systems, if the information about these attacks were shared, defenders can adapt their defense as quickly as the attackers offense. By creating an optimal social network that incentivizes sharing of this information we can decrease the number of effective attacks on the members of the network. This network will ideally optimize implementation costs for all active participants while simultaneously strengthening their security against novel attacks. Our approach to designing this network will be shown with an Agent Based Model (ABM) in a public goods game (PGG) that accounts for the possible actions of both defenders and attackers. The design will attempt to maximize sharing among defenders, minimize costs incurred by defenders, and reduce the total number of successful attacks.

1. Model

- 1.1. The goal of this model is to simulate a social network of defenders utilizing cybersecurity strategies as a public good. By manipulating certain variables of this network such as: number of groups, number of members per group, punishment cost to defender, punishment cost to free-riders, probability of adding a friend, and the probability of removing a friend we hope to maximize the attributes: reciprocity, Probability of contribution, value of contribution, defender altruism, number of friends per defender, return on security investment, and number of attempted attacks mitigated. Finally we aim to minimize: number of successful attacks, implementation costs for countermeasures, percentage of free-riders, and return on attack for attackers.

1.2. Variables:

- 1.2.1. number of groups: q
1.2.2. punishment cost: $\gamma_{i,j}$
1.2.3. probability of adding a friend: $\varphi = C_i$
1.2.4. probability of removing a friend: $\psi = \frac{\sum_i c_i}{\varphi_i}$
1.2.5. Reciprocity: $\xi = f_{\text{contribution}} \times f_{\text{contribution value}}$

- 1.2.6. Probability of Contribution [3]: $c_i = \frac{1}{1 + e^{\frac{\pi_i^t - \pi_i^{t-1}}{k}}}$
1.2.7. Value of Contribution: $cv_i = \text{Security investment} \times f_{\text{RM}}$

1.2.8. Altruism[3]:

$$\pi_i(g_i, g_j) = g_i - \alpha_i \max[g_j - g_i, 0] - \beta_i \max[g_i - g_j, 0]$$

1.2.9. Number of friends per defender: $x \subset N: 0 \leq x \leq |\text{graph}|$

1.2.10. Return on security investment [1]: $ROSI = \frac{ben_{ac} - CSI_c}{CSI_c}$

1.2.11. Number of attempted attacks mitigated[1]: am_i

1.2.12. Implementation costs: $\sum CM(SI)$

1.2.13. We aim to show with our model that given: $0 < q, \gamma_{i \in 0,1}, c_i = 1, \varphi$. That $f(q, \gamma_i, c_i, \varphi)$ will maximize the attributes reciprocity, probability of contribution, value of contribution, friends per defender, return on security investment, attacks mitigated and result in the optimal network design for sharing cybersecurity information.

1.3. The network we used is an unweighted bidirectional planted partition graph to represent a social network. This graph is divided into X cliques and Y members per clique. Initially, each clique has c probability of a connection within the group and v probability of a connection outside the group. Each node on the graph is used to represent a defender agent that can communicate with any defender it shares an edge with. All defenders can remove edges at will and create edges with approval of the connecting defender. Edge removal will be referred to as friend removal for the sake of clarity and is determined by

$$remove\ friend = \frac{\sum c_i}{\# of\ contributions} < v \text{ Where } \sum c_i \text{ is the sum of the value of all}$$

contributions. The defender will remove a friend whose average contribution value falls below a variable threshold, v. Defenders will be added from a request list if they meet the criteria defined by $add\ friend = \pi > 100 \ \&\& \ c_i > 0.6$. Defenders will request to establish a friendship if they see an agent meets the

criteria defined by $request\ friend = \frac{\sum c_i}{\# of\ contributions} > v$. Defenders have access to the contribution history and altruistic tendencies of all other defenders in the network. Defenders have access to the shared countermeasures and attempted attacks on friends starting from the beginning of the friendship.

1.4. Throughout the entire run defenders are susceptible to have a set of attacks attempted upon them. Attack sets are determined by choosing a combination of attacks that will optimize return on attack (ROA) determined by (1):

$$ROA_{AC} = \frac{GI - (GI \times RM_{AC}) - cost_A}{cost_A} \text{ where } GI \text{ is the expected gain of the attack, a, } RM_{AC}$$

is the risk mitigated by the set C when is used against the set A and is computed

as $RM_{AC} = \min_{a \in A} (RM_{AC})$ and $cost_A = \sum_{a \in A} (cost_A)$. C is determined by the most effective countermeasure against the specified attack. Once the best attack set is chosen, each attack has a chance to be attempted, shown in table 1, and then a separate chance to be successful based on the countermeasure set used by the defender. All attempted attacks are stored in a list to be potentially shared with all befriended defenders. If an attack is successful then a predetermined amount of wealth is subtracted from the defender.

- 1.5.** The key component of this model is that defenders can choose to share information amongst themselves and use that information to determine the best set of countermeasures with the end goal of maximizing sharing in the network and reducing the overall number of successful attacks. Our sharing model is based on [2] using social preferences models with risk aversion that break down into two main elements of self-regarding and other-regarding preferences. we express our utility function (2) as below:

$$\pi_i(g_i, g_j) = g_i - \alpha_i \max[g_j - g_i, 0] - \beta_i \max[g_i - g_j, 0]$$

where α_i and β_i represent the constant elasticity of substitution in this function to exhibit the elasticity of the ratio of other-regarding preferences and individualism, respectively. Equation 4 results in a the probability to contribute, κ , where $0 \leq \kappa \leq 1$, If the number of freeriders is greater than 0, otherwise $\kappa = 1$.

$$\frac{1}{1 + e^{\frac{\pi_i^t - \pi_i^{t-1}}{k}}}$$

If the defender chooses to contribute during the round then it will share with all connected nodes the set of countermeasures it used to defend against any attacks and a set of successful or attempted attacks that happened to it during the round. These contributions are given a monetary value based on their cost to implement and the risk mitigation they provide as shown in table 3, for countermeasures, and the possible loss for the value of attacks in table 2. The combined value of these contributions are then used to determine by other defenders if the history of contributions is equitable compared to their own history of contributions. If the disparity between the two is too great, defenders will terminate the connection between them, as shown in section 1.2.

- 1.6.** Each round defenders decide on the best defense strategy by predicting what possible attacks will happen and then deciding the best set of countermeasures to use that will give the highest return on security investment (ROSI). Using the

strategy for multiple attack and countermeasure scenarios in [1] each defender determines its ROSI from equation 3:

$$ROSI = \frac{ben_{ac} - CSI_c}{CSI_c} \quad (3)$$

Where ben_{ac} is shown in eq. 2 as the benefit produced by the set C used to stop the set of attacks A and CSI_c is the total cost associated to the set of

countermeasures C: $CSI_c = \sum_{c \in C} CSI_c$.

$$ben_{AC} = \min\{AV, \sum_{a \in A} [ALE_a - ALE_a \times (1 - RM_{ac})]\} \quad (4)$$

ALE is shown in eq. 3 as the annual loss expectancy in a year from attack a, RM_{ac} is the risk mitigated by the set of countermeasures C on attack a, and AV is the asset value represented in monetary units.

$$ALE = (Number\ of\ Incidents\ per\ Year) \times (Potential\ Loss\ per\ Incident) \quad (5)$$

The ALE is determined initially based on a random guess at which attacks from table 1 a defender could face but can be updated each round if a neighbor contributes a set of attacks that happened to them. Each defender has access to all possible combinations of countermeasures and chooses a set which gives the highest ROSI with a given predicted attack set.

- 1.7.** In this model, we implement a decentralized punishment strategy by contributors with the goal of increasing the average level of contribution by the Defenders. (6) gives the probability of punishing j by i if the contribution of i is more than a defined threshold. The punishment would be carried out by i if $P_{ij} = 1$.

$$p_{ij} = \min \left\{ \frac{|\arctan(\beta, \alpha)|}{2\gamma_i} \lambda_j - \gamma_i, 1 \right\} \quad (6)$$

Agent i chooses one of the free-riders proportionally to their payoff and if a free-rider's average contribution is below a variable percentage, g. In our model, this obeys from the Moran rule where probability of choosing agent is given by

$$Prob(punishing\ j) = \frac{\pi_j}{\sum_{l=1}^N \pi_l} \quad (7)$$

Measures

Reciprocity

Notation	Meaning
g_i	Monetary gain of the defender i
c_i	Contribution of defender i to provide cybersecurity
C_i	Average contribution
cv_i	Monetary value of contribution
γ_i	The cost incurred by the punisher j
λ_i	Penalty of punishing i
R	Resource of agents
P_i	Probability of punishing j by i
P_i	Probability of punishing i by j
π	Altruism
CM	Set of countermeasures
q	Number of cliques
m	Number of members per clique
φ	Probability of adding a friend
ψ	Probability of removing a friend
x_i	Number of friends per defender i
ξ	reciprocity
am_i	Number of attacks mitigated by defender
Defenders	All the agents that belong to the defense group
defender	An agent that is a Defender

Attackers	All the agents that belong to the offense group
attacker	An agent that is an Attacker

Attack	Probability of Attack	Attack Cost	Attack Impact
A1: Malware	0.25	50	2.6
A2: Web-based attacks	0.20	60	2.3
A3: Denial of service	0.20	70	1.7
A4: Malicious insider	0.15	65	1.6

Security Measures	Security Investment	Annual Cost	RM1	RM2	RM3	RM4
CM1: Security intelligence and threat sharing	100	25	0.6	0.5	0.4	0.5
CM2: Advanced identity and access management	80	30	0.4	0.6	0.4	0.6
CM3: Cyber and user behavior analytics	110	30	0.5	0.5	0.4	0.6
CM4: Cryptography technologies	100	5	0.4	0.5	0.3	0.4
CM5: Automated policy management	80	45	0.5	0.4	0.3	0.5
CM6: Enterprise governance, risk, and compliance	300	50	0.5	0.5	0.4	0.5

- [1] [Evaluation of complex security scenarios using defense trees and economic indexes](#)
- [2] [Social Preferences in Decision Making Under Cybersecurity Risks and Uncertainties](#)
- [3] [Advancing the concept of cybersecurity as a public good](#)
- [4] [Friend recommendation with content spread enhancement in social networks](#)
- [5] [The Analysis of Social Networks](#)